CMU 11-781 / 02-741 GenAI for Biomedicine – Homework 2

Sri Datta Bandreddi
sbandred@andrew.cmu.edu

Setup. All analyses were run by an AI agent workspace (Claude Code, model Claude Opus 5.5) on a CPU-only Linux machine; no GPU or structure prediction was needed. All code and outputs are in the submitted `submission/` folder; `bash submission/run_all.sh` rebuilds the environment, downloads and checksums the data, reruns every analysis and compiles this PDF in about 4 minutes. Claims are tagged [OBS] (computed here), [PRED] (interpretation) or [EXT] (retrieved, with citation).

Checkpoint evidence index. All checkpoints: *complete* (A: complete once the student-verification boxes are filled in).

ID	Evidence	ID	Evidence
P1	1(a), p. 1; Table 1; `protein/code/01`	S5	2(e), p. 5; Fig. 3; `single_cell/code/04`
P2	1(b), p. 2; Table 2; `protein/code/01--02`	T1	3(a), p. 6; Fig. 4; `spatial/code/01`
P3	1(c), p. 2; Table 3	T2	3(b), p. 7; Table 7; `spatial/code/02`
P4	1(d), p. 3; Fig. 1, Table 3; `protein/code/03`	T3	3(c), p. 7; Fig. 5; `spatial/code/03`
S1	2(a), p. 4; Table 4, Fig. 2; `single_cell/code/01`	T4	3(d), p. 8; Table 6; `spatial/code/03`
S2	2(b), p. 4; Table 4	T5	3(e), p. 8; Fig. 6, Table 7; `spatial/code/02,04`
S3	2(c), p. 4; Table 5; `single_cell/code/02`	A	1(e), 2(f), 3(f)
S4	2(d), p. 4; Table 5, Fig. 2; `single_cell/code/03`	R	`README.md`, `run_all.sh`, `requirements.txt`; final section

Problem 1 – Predicting the effects of EGFR mutations (G719S, T790M, L858R)

(a) Map the variants (P1). *Identify the protein, verify each variant against its sequence, and map all candidate residues to the chosen structure.*

The protein is human EGFR, UniProt P00533 (1,210 aa, entry v301); all three variants lie in the kinase domain (712–979) [EXT]. The stated wild-type residues match UniProt at G719 (first Gly of the GxGxxG P-loop), T790 (gatekeeper) and L858 (three residues after DFG 855, in the activation segment) [OBS].

Each structure's full SEQRES was aligned to UniProt (Biopython, BLOSUM62) and cross-checked against the depositors' SEQADV records (Table 1). The key finding is that **4HJO uses mature-protein numbering, canonical** −24: L858 is Leu834 and T790 is Thr766 in 4HJO. In every structure the only mismatches against UniProt are the deposited mutations, including 4HJO's engineered V948R, which keeps the kinase monomeric and inactive [19] and lies 26–31 Å from all three sites. AlphaFold [4] pLDDT is 73 (G719), 88 (T790) and only 51 (L858), consistent with a flexible activation loop.

Table 1: Residue mapping (`table_p1b`). Offset = author − canonical numbering.

PDB	Content	Offset	G719 / T790 / L858	Mismatches vs UniProt
4HJO	WT+V948R, erlotinib, inactive [11]	−24	Gly695 / Thr766 / Leu834	R948 (engineered)
1M17	WT, erlotinib, active-like [13]	−24	Gly695 / Thr766 / Leu834	none
2ITX / 2ITN / 2ITV	WT/G719S/L858R + AMP-PNP [17]	0	canonical	none / S719 / R858
2JIT	T790M, apo [18]	0	Gly / **Met** / Leu	M790
4I22	L858R/T790M/V948R, gefitinib [2]	0	Gly / **Met** / **Arg**	M790, R858, R948

(b) Measure structural evidence (P2). *Define and report reproducible measurements for each candidate.*

Definitions: heavy atoms only; distance = minimum inter-atomic distance; neighbours = residues within 5 Å; relative SASA from freesasa (1.4 Å probe, ligands removed); αC state from K745 NZ–E762 OE (< 4 Å = αC-in). Results in 4HJO (Table 2) [OBS]:

- **T790** is closest to erlotinib (3.48 Å), and is the closest of the three sites to the ligand in all six ligand complexes.
- **L858** is buried (SASA 0.10, vs 0.15–0.40 in αC-in structures), 2.96 Å from D855 and 4.57 Å from αC residue M766, in a mostly hydrophobic shell that also contains K745, D855 and K860.
- **G719** is exposed (SASA 0.35) and 5.06 Å from erlotinib. In G719S (2ITN), Ser OG is 5.15 Å from the nearest phosphate (no direct H-bond), and the P-loop moves $\leq$1.95 Å.
- **Met790 check:** apo Met790 (2JIT) superposed on 4HJO appears to clash with erlotinib (0.6–1.3 Å), but drug-bound Met790 (4I22) sits 3.3–3.5 Å from the drug. The clash comes from the apo rotamer (in the active-like 1M17 pocket it is a 2.5 Å near-contact).

Table 2: Site measurements, Å (4HJO, K745–E762 = 13.3 Å, αC-out; 8 structures in `table_p2a`).

Site	→ erlotinib	→ K745 NZ	→ E762	→ D855	→ αC helix	rel. SASA
G719	5.06	7.78	20.35	10.02	16.5	0.35
T790	**3.48**	9.03	14.27	8.57	7.5	0.21
L858	5.26	3.58	6.18	**2.96**	**4.57**	**0.10**

(c) Interpret with evidence (P3). *Compare candidate mechanisms using observed results and cited sources; separate observations from predictions; state uncertainty.*

Table 3 compares the three variants. Ranked by how directly the structures support a mechanism: **L858R** > **T790M** > **G719S**. A single impact ranking is not meaningful, because L858R and G719S are activating mutations that predict inhibitor sensitivity [7, 8], while T790M causes acquired resistance [5, 10]. AlphaMissense [1] scores all three $\geq$0.96 and cannot separate these classes. Main uncertainties: mechanisms are inferred from static structures that carry V948R, and 4I22 (itself αC-out) accommodates Arg858 via salt bridges to D855/D837, which argues against a simple incompatibility model for L858R.

Table 3: Comparison of the three variants (all measurements from `table_p2a/d/e`).

Variant	Structural evidence [OBS]	Mechanism [PRED] / support [EXT]	Class	Confidence
L858R	Buried in inactive A-loop helix (SASA 0.10); packs against αC M766 (4.57 Å); pLDDT 51	Arg disrupts the pocket holding the inactive A-loop against αC, shifting the balance to the active state [17, 12]; 4I22 shows a shift, not an incompatibility	Activating	Mod.–high
T790M	Gatekeeper, 3.48 Å to erlotinib; drug-bound Met does not clash	Resistance mainly via higher ATP affinity [18]; no evidence for a purely steric block	Resistance	Moderate
G719S	Exposed P-loop Gly; Ser OG 5.15 Å to phosphate; P-loop shift $\leq$1.95 Å	Altered P-loop flexibility destabilises the inactive state [17]; no new contact observed	Activating	Low–mod.

(d) Communicate the comparison (P4). *Provide an annotated EGFR structure and one table covering all three variants.*

Figure 1 shows the annotated structure; Table 3 is the comparison table.

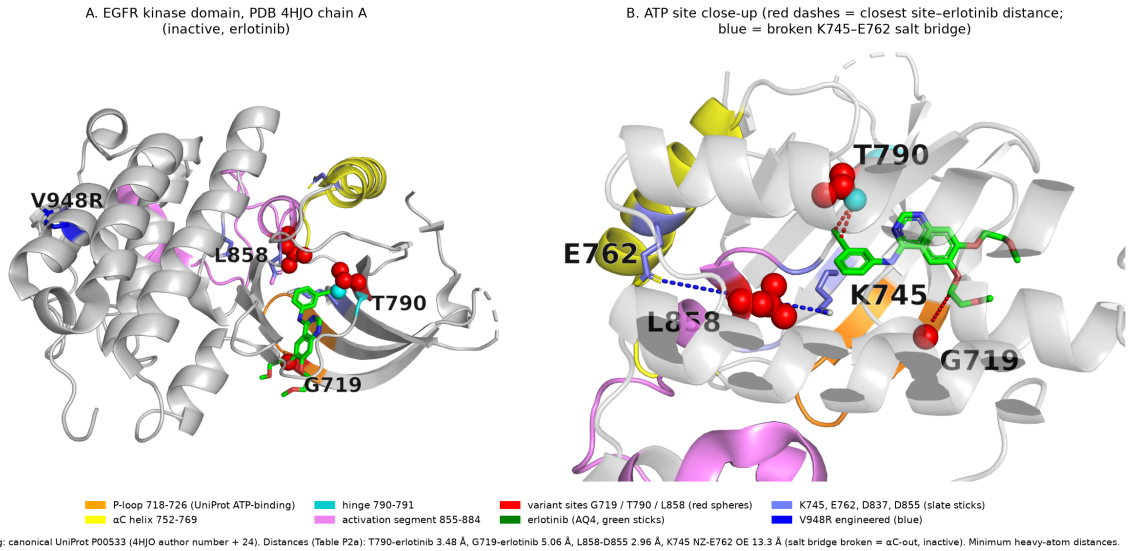


Figure 1: EGFR kinase domain, PDB 4HJO, canonical numbering. (A) Overview; (B) ATP site. Red spheres: variant sites; green: erlotinib; red dashes: closest site–erlotinib distances; blue dashes: broken K745–E762 salt bridge (αC-out). Interactive: `egfr_4hjo_interactive.html`.

(e) Agent audit. **Sources:** UniProt P00533 (+ paralogs P04626, P21860, Q15303); PDB 4HJO, 1M17, 2ITX, 2ITN, 2ITV, 2JIT, 4I22; AlphaFold DB AF-P00533-F1 v6 (model, pLDDT, AlphaMissense); 13 papers. **Computed:** sequence checks, mapping, distances, SASA, αC state, superpositions, paralog alignment. **Retrieved:** sequences, annotations, coordinates, SEQADV records, pLDDT, AlphaMissense, published mechanisms. **Independent checks:** (1) mapping agrees with every SEQADV record (4HJO lists Arg924 as engineered V948, confirming −24); (2) all 13 citations resolve on PubMed (this caught one wrong PMID, now fixed). **Weaknesses found:**

a mapping bug gave an inconsistent offset for 2ITX/2ITV (fixed by aligning full SEQRES); a naive Met790 superposition falsely suggested a steric clash (refuted with 4I22); a review found L858's charged neighbours and the 4I22 counter-evidence had been left out (now reported, confidence lowered).

Student verification (to complete before submission): open 4HJO at RCSB, confirm Leu834 = canonical L858, measure Thr766–erlotinib in Mol* and compare with 3.48 Å. Result: _____

Problem 2 – Single-cell cell types and robustness (PBMC3k)

(a) Baseline and QC (S1). *Report QC decisions, settings and seed, cluster sizes, and a cluster-labelled UMAP.*

Data: 10x PBMC3k (2,700 cells). QC kept cells with 200–2,500 genes and $< 5\%$ mitochondrial reads (removes 5 + 57 cells; medians 817 genes, 2.03% mito) and genes in ≥ 3 cells, leaving **2,638 cells** $\times$ **13,656 genes**. Settings: normalise to 10^4, log1p, 2,000 HVGs, regress out counts and % mito, scale, PCA (30 PCs), kNN $k = 15$, Leiden resolution 1.0, UMAP; **seed 0** everywhere [16, 14]. This gives 9 clusters (sizes in Table 4; UMAP in Fig. 2A). Resolution 1.0 was chosen after seeing that 0.6 merges CD8 T and NK cells; this is disclosed as a post-hoc choice.

(b) Cell-type evidence (S2). *Use markers computed from your PBMC3k data and cited biology to support major labels.*

Labels come from Wilcoxon markers computed on these data, plus a rule that scores literature marker sets per cluster (ambiguous if the top two differ by < 0.15). No reference labels were used.

Table 4: Clusters, data-derived markers and labels (`table_s2b/d`; marker biology from [20, 3, 15]).

Cl.	Cells	Top markers (Wilcoxon, these data)	Label
0	508	IL32, LTB, CD3D, IL7R, S100A4	Memory CD4 T
1	643	ribosomal genes, LDHB, CD3D; CCR7 in 44%	Naive T, CD4/CD8 mixed (see 2(e))
2	342	CD79A, MS4A1, CD79B	B
3	157	FCGR3A, LST1, AIF1	FCGR3A$^+$ monocyte
4	155	GNLY, GZMB, NKG7, PRF1	NK
5	306	CCL5, NKG7, GZMK, CD3D; CD8A in 51%	CD8 T
6	475	S100A8, S100A9, LYZ, FCN1	CD14$^+$ monocyte
7	38	FCER1A, CST3, HLA class II	Dendritic cell
8	14	PPBP, PF4	Platelet

(c) Resolution check (S3). *Compare at least three Leiden resolutions with other settings fixed; quantify splits/merges and discuss label stability.*

Leiden at 0.3, 0.6, 1.0 and 1.4 on the same graph (Table 5). **B cells** are perfectly stable (recall and purity 1.00). NK and CD14 monocyte cells always keep their labels, but at low resolution their clusters absorb other cells (the "NK" cluster is 62–64% CD8 T at 0.3–0.6). CD8 T, naive and memory CD4 exist only at ≥ 1.0, and at 1.4 the naive/memory boundary moves again. Changing only the seed gives ARI 0.77–0.83, mostly due to Leiden's 2-iteration default (0.94 when run to convergence); almost all of that change is at the naive/memory boundary.

(d) Downsampling check (S4). *Rerun a fixed-seed 60–80% sample and compare retained cells quantitatively.*

A 70% sample (1,847 cells, seed 42) re-run from raw counts gives 7 clusters, ARI 0.65 and **77.6% label agreement** on shared cells (Fig. 2B). B, NK and monocytes agree at ≥0.95. CD8 T cells split between NK (55%) and memory CD4 (43%), and the 8 platelets join the FCGR3A monocytes. Re-clustering at 1.2, or using five other subsample seeds, recovers CD8 T every time; platelets return in only 1 of 5. So the CD8/NK split is real but near the resolution threshold, and 14 platelets are too few to survive subsampling.

Table 5: Resolution and downsampling checks (`table_s3*`, `table_s4*`).

Comparison	Clusters	ARI	NMI	Main change
Resolution 0.3 vs 1.0	5	0.59	0.77	T, NK/CD8 and myeloid clusters merge
Resolution 0.6 vs 1.0	7	0.64	0.82	CD8 T + NK; naive + memory CD4
Resolution 1.4 vs 1.0	11	0.70	0.80	naive/memory re-split (72% / 82% kept)
Leiden seeds 1–5	9	0.77–0.83	–	naive/memory boundary moves
70% subsample, seed 42	7	0.65	0.75	CD8 T and platelets lost
Subsample seeds 1–5	8–9	–	–	CD8 T 5/5, platelets 1/5

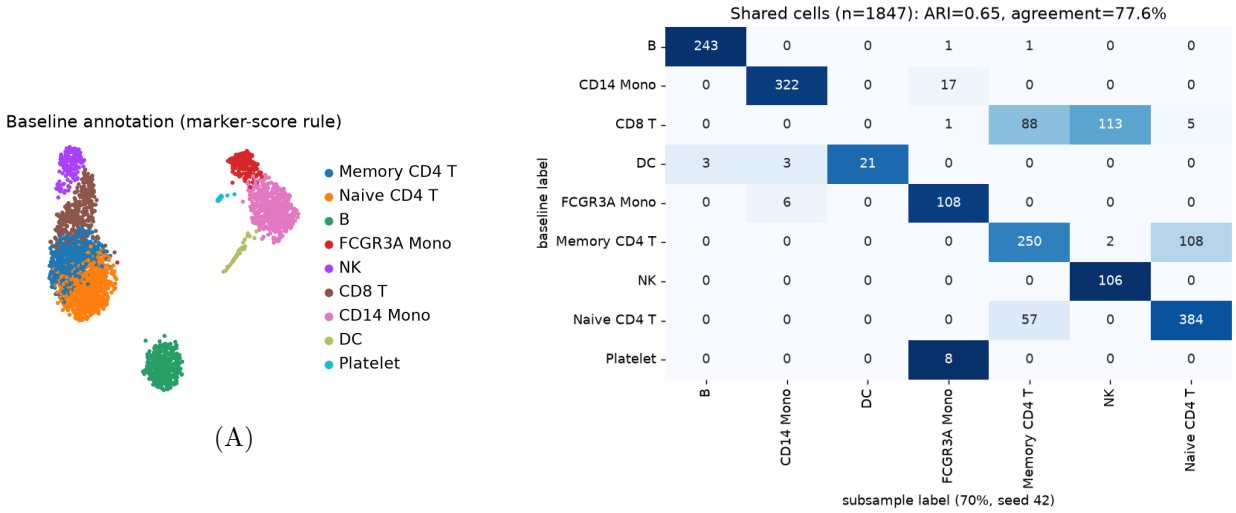


Figure 2: PBMC3k. (A) Baseline UMAP with labels (seed 0, resolution 1.0). (B) Baseline vs 70% subsample labels on shared cells.

(e) Ambiguous population (S5). *Investigate one unclear population with marker comparisons and a reason for the uncertainty.*

Naive vs memory CD4 (clusters 1 and 0). The two differ only modestly (cluster 0: S100A4, S100A11, IL32 higher; cluster 1: CCR7, SELL and ribosomal genes higher). The naive-minus-memory score is not bimodal (bimodality coefficient 0.39 < 0.555; AUC 0.73), and 16–21% of each cluster's kNN edge weight points to the other cluster (Fig. 3) [OBS]. *Reason* [PRED]: naive → central memory → effector memory is a continuous differentiation path, so the markers form a gradient rather than a gap. Cluster 1 also contains CD8 cells: 26% express CD8A/B, against 3–10% background in B/NK cells and only 60% detection in true CD8 T, giving an estimated 31–40% CD8 T cells. Naive CD8 cells share the CCR7/SELL/LEF1 programme, and CD4 mRNA is detected

in only 9–12% of CD4 T cells, so we relabel cluster 1 "Naive T, CD4/CD8 mixed". Protein data (CITE-seq) would be needed to resolve it.

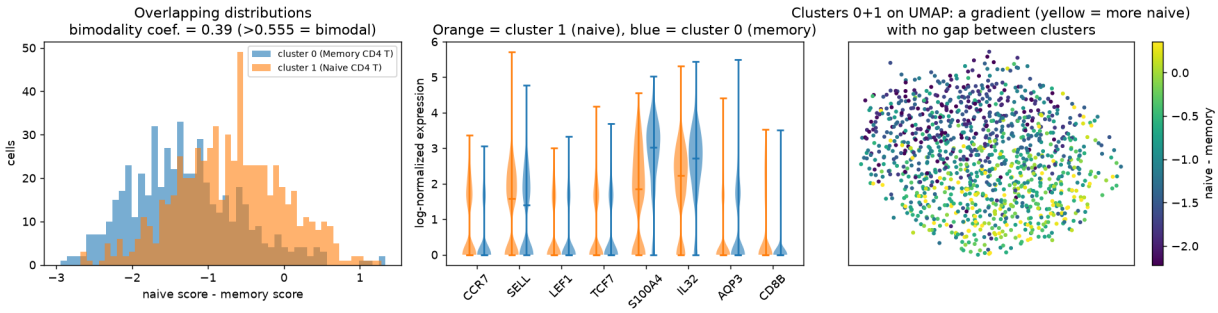


Figure 3: Naive vs memory CD4. Left: overlapping score distributions. Middle: graded markers (orange = cluster 1, blue = cluster 0). Right: the score on the UMAP, a gradient without a gap.

(f) Agent audit. **Sources:** PBMC3k matrix (10x Genomics via course repository, SHA-256 checked); marker biology [20, 3, 15]; scanpy [16], Leiden [14]. **Computed:** QC, clustering, UMAP, Wilcoxon markers, marker-set scores, ARI/NMI, subsampling, seed sweeps, bimodality, CD8 detection rates. **Retrieved:** canonical marker lists and cell-type biology. **Independent checks:** (1) seed-only control (ARI 0.77–0.83; 0.94 at convergence) calibrates the resolution and subsample effects; (2) the CD8 loss was re-tested at resolution 1.2 and with five subsample seeds and recovered every time, so it is a threshold effect. **Weaknesses found:** the automatic rule mislabelled cluster 1 as "Naive CD4 T" (relabelled; a first fix ignored CD8 dropout and was corrected after review); the resolution was chosen post hoc; recall-only stability hid impure NK clusters (purity added); Fig. 3 first had reversed colours and a saturated scale (fixed).

Student verification (to complete before submission): from `table_s2b_top10_markers.csv` alone, assign clusters 2, 4 and 6; rerun `03_downsample.py` and confirm 77.6% agreement. Result: _____

Problem 3 – Spatial transcriptomics (Visium adult mouse brain)

(a) Expression-only domains (T1). *Cluster without spatial information; show the same labels in expression space and on tissue, with settings and spot counts.*

QC (≥5,000 UMIs, ≥1,000 genes, < 30% mito) kept **2,683 of 2,702 spots** (17,535 genes). Clustering used an object built from counts only; assertions check that no coordinates or image are present. Settings: seurat_v3 HVGs (2,000), log-normalise, PCA (30), kNN $k = 15$, Leiden resolution 0.5, seed 0. Result: **16 domains** (20–339 spots), shown in Fig. 4. Tentative anatomy from markers and the Allen atlas [6] [PRED]: cortical layers (2, 3, 15), thalamus (4, 5), fibre tracts (8), hippocampus (11–13), hypothalamus (6), ventricle (14), meninges (9), striatum-like (1); domains 0, 7 and 10 are uncertain.

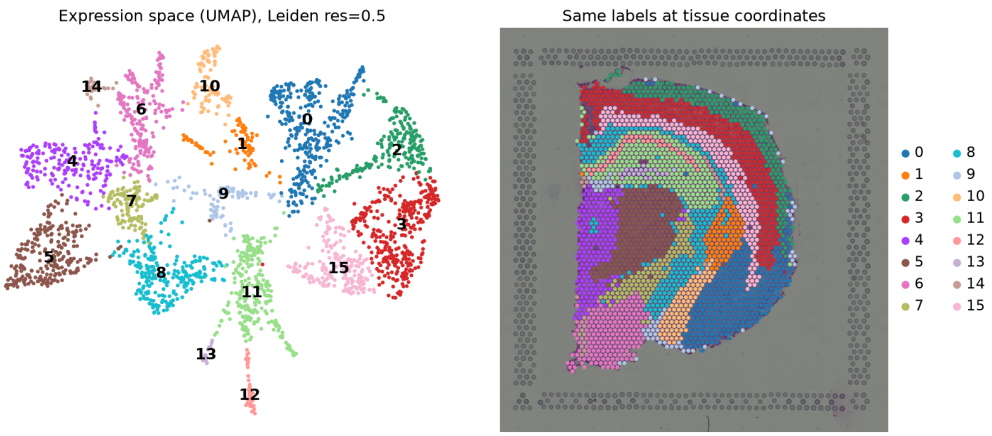


Figure 4: Expression-only domains: the same labels and colours in UMAP space (left) and on tissue (right).

(b) Spatial organisation (T2). *Define the physical-neighbour graph and report an interpreted numerical spatial statistic.*

Graph: the Visium hexagonal lattice, each spot linked to its up-to-6 adjacent spots ($\sim$100 μm apart; squidpy [9]), 7,759 edges. **Statistic:** the fraction of edges joining two spots of the same domain. Observed **0.783**, against 0.084 expected by chance (Table 7): most physical neighbours share an expression domain. Every domain is 10–98$\times$ more self-adjacent than chance, and the strongest between-domain contacts are between hippocampal sub-fields. This shows spatial coherence, not mechanism.

(c) Spatial genes (T3). *State the tested gene set and method, rank spatially variable genes, and map at least two on tissue.*

Gene set: the top 200 HVGs (no mitochondrial genes). Method: Moran's I on the same graph, BH FDR. All 200 are significant, which is expected for pre-selected genes, so we use ranks. Top: Prkcd (I = 0.87, thalamus), Mbp (fibre tracts), Ttr (choroid plexus), Pmch; lowest: blood and vascular genes (Ccl5, S100a8/9). Five genes are mapped in Fig. 5.

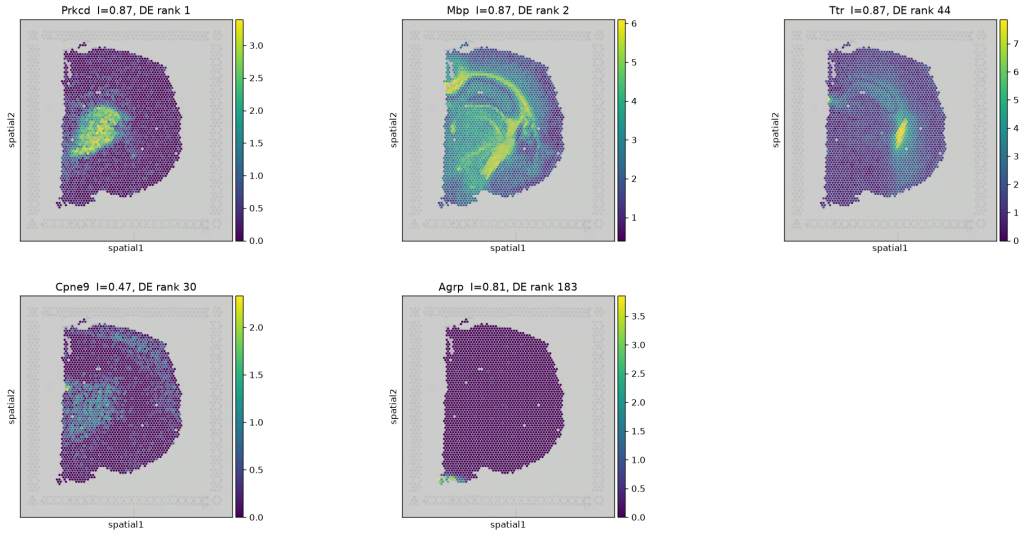


Figure 5: Selected genes on tissue, with Moran's I and DE rank.

(d) Compare gene rankings (T4). *Compare DE and spatial rankings for the same genes, with examples that explain their differences.*

DE: Wilcoxon one-vs-rest across domains on the same 200 genes. Spearman ρ between Moran's I and DE z is **0.52**, falling to 0.29 if DE is ranked by fold change. Examples (Table 6): Prkcd and Mbp mark large regions that match a domain, so they rank high in both. *Agrp* is spatially sharp (Moran rank 7) but DE rank 183: it is confined to the small arcuate region, so most spots of its best domain are zero and the rank-based test penalises it, despite a large fold change ($\log_2$FC 5.5). Cpne9 separates domains on average but varies in a graded way inside them. Moran's I measures neighbour agreement; DE measures separation of predefined groups.

Table 6: Ranking examples (rank of 200; `table_t3b_t4_moran_and_DE_rankings.csv`).

Category	Gene	Moran rank	DE rank (z)	DE rank ($\log_2$FC)	% spots detected
High in both	Prkcd / Mbp	1 / 2	1 / 2	41 / 102	53 / 100
Spatial-high, DE-low	Agrp	7	183	27	4
	Fezf1	22	161	62	5
DE-high, spatial-lower	Cpne9	103	30	171	42

(e) Negative control (T5). *Use ≥100 fixed-seed randomisations; compare the same observed and null statistic; report an empirical comparison and a limitation.*

Null: domain labels permuted across spots on the fixed graph, **1,000 times, seed 0**, keeping domain sizes and the graph. Empirical $p = (1+\#\{\text{null} \geq \text{obs}\})/(N+1)$, one-sided. No permutation came close: null 0.084 ± 0.003 vs observed 0.783 ($z = 224$, $p = 0.001$, the minimum possible; Fig. 6). A second null that shuffles positions and rebuilds the graph (seed 1) gives the same result and serves as an implementation check. Because depth and % mito are themselves spatial (Moran's I 0.49, 0.58), we regressed both out and re-clustered: 13 domains, ARI 0.78 with the originals, still coherent (0.792 vs 0.093). **Limitation:** the null assumes complete spatial randomness, a weak alternative that any smooth gradient would beat; a null preserving autocorrelation was not run. Visium spots also mix cells and share diffused transcripts.

Table 7: Observed vs null (`table_t5a`, `table_t5c`).

Statistic	Observed	Null mean ± SD	z	Empirical p
Same-domain edge fraction (16 domains)	0.783	0.084 ± 0.003	224	0.001
After depth/mito regression (13 domains)	0.792	0.093 ± 0.003	–	0.001

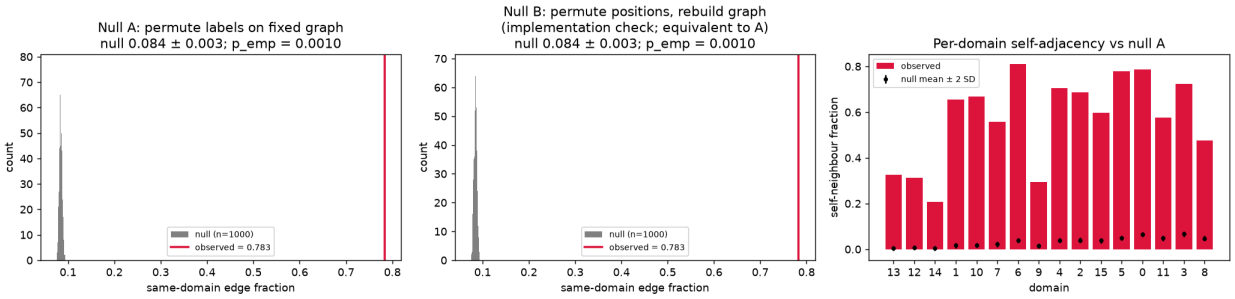


Figure 6: Observed statistic (red) vs 1,000 permutations. Left: label permutation; middle: graph-rebuild check; right: per-domain self-adjacency vs null mean ±2 SD.

(f) Agent audit. **Sources:** 10x Genomics V1 adult mouse brain matrix and spatial archive (course repository, SHA-256 checked); squidpy [9]; Allen atlas [6]. **Computed:** QC, domains, graph, same-domain fraction, neighbourhood enrichment, Moran's I, DE, rank correlations, permutation nulls, depth/mito regression. **Retrieved:** anatomical meaning of marker genes. **Independent checks:** (1) the permutation mean (0.0844) equals the analytic expectation $\sum_k n_k(n_k-1)/(N(N-1)) = 0.0844$; (2) a reviewer rebuilt the hexagonal graph by hand and got the same 7,759 edges and statistic. **Weaknesses found:** the second null first rebuilt its graph from array coordinates, which kept only 4 neighbours per spot, so it was compared on a different graph (fixed; it now asserts identical edges); only the z-based DE ranking was reported at first (fold-change ranking added); the null is weak; anatomy is not checked against atlas images.

Student verification (to complete before submission): compare two domains in Fig. 4 with the Allen Brain Atlas coronal reference and their markers in `table_t1c`. Result: _____

Reproducibility

Run `bash submission/run_all.sh` (order and outputs in `submission/README.md`). Python 3.12.3 via `uv`; exact versions in `requirements.txt` (scanpy 1.12.5, squidpy 1.8.4, leidenalg 0.12.0, scikit-learn 1.9.1, gemmi 0.7.5, Biopython 1.88, freesasa 2.2.1, PyMOL 3.2). Seeds: single cell 0 (subsample 42, extra seeds 1–5); spatial 0 (null) and 1 (graph-rebuild check); protein deterministic. Data: 10x PBMC3k and V1_Adult_Mouse_Brain (course copies, checksummed); UniProt P00533; PDB 4HJO, 1M17, 2ITX, 2ITN, 2ITV, 2JIT, 4I22; AlphaFold DB AF-P00533-F1 v6.

References

[1] Cheng J, et al. Accurate proteome-wide missense variant effect prediction with AlphaMissense. *Science* 381:eadg7492, 2023. PMID 37733863.

[2] Gajiwala KS, et al. Insights into the aberrant activity of mutant EGFR kinase domain and drug recognition. *Structure* 21:209–219, 2013. PMID 23273428.

[3] Hao Y, et al. Integrated analysis of multimodal single-cell data. *Cell* 184:3573, 2021. PMID 34062119.

[4] Jumper J, et al. Highly accurate protein structure prediction with AlphaFold. *Nature* 596:583–589, 2021. PMID 34265844.

[5] Kobayashi S, et al. EGFR mutation and resistance of non-small-cell lung cancer to gefitinib. *N Engl J Med* 352:786–792, 2005. PMID 15728811.

[6] Lein ES, et al. Genome-wide atlas of gene expression in the adult mouse brain. *Nature* 445:168–176, 2007. PMID 17151600.

[7] Lynch TJ, et al. Activating mutations in the EGFR underlying responsiveness of NSCLC to gefitinib. *N Engl J Med* 350:2129–2139, 2004. PMID 15118073.

[8] Paez JG, et al. EGFR mutations in lung cancer: correlation with clinical response to gefitinib therapy. *Science* 304:1497–1500, 2004. PMID 15118125.

[9] Palla G, et al. Squidpy: a scalable framework for spatial omics analysis. *Nat Methods* 19:171–178, 2022. PMID 35102346.

[10] Pao W, et al. Acquired resistance of lung adenocarcinomas to gefitinib or erlotinib is associated with a second mutation in the EGFR kinase domain. *PLoS Med* 2:e73, 2005. PMID 15737014.

[11] Park JH, Liu Y, Lemmon MA, Radhakrishnan R. Erlotinib binds both inactive and active conformations of the EGFR tyrosine kinase domain. *Biochem J* 448:417–423, 2012. PMID 23101586.

[12] Shan Y, et al. Oncogenic mutations counteract intrinsic disorder in the EGFR kinase and promote receptor dimerization. *Cell* 149:860–870, 2012. PMID 22579287.

[13] Stamos J, Sliwkowski MX, Eigenbrot C. Structure of the EGFR kinase domain alone and in complex with a 4-anilinoquinazoline inhibitor. *J Biol Chem* 277:46265–46272, 2002. PMID 12196540.

[14] Traag VA, Waltman L, van Eck NJ. From Louvain to Leiden: guaranteeing well-connected communities. *Sci Rep* 9:5233, 2019. PMID 30914743.

[15] Villani AC, et al. Single-cell RNA-seq reveals new types of human blood dendritic cells, monocytes, and progenitors. *Science* 356:eaah4573, 2017. PMID 28428369.

[16] Wolf FA, Angerer P, Theis FJ. SCANPY: large-scale single-cell gene expression data analysis. *Genome Biol* 19:15, 2018. PMID 29409532.

[17] Yun CH, et al. Structures of lung cancer-derived EGFR mutants and inhibitor complexes: mechanism of activation and insights into differential inhibitor sensitivity. *Cancer Cell* 11:217–227, 2007. PMID 17349580.

[18] Yun CH, et al. The T790M mutation in EGFR kinase causes drug resistance by increasing the affinity for ATP. *PNAS* 105:2070–2075, 2008. PMID 18227510.

[19] Zhang X, et al. An allosteric mechanism for activation of the kinase domain of epidermal growth factor receptor. *Cell* 125:1137–1149, 2006. PMID 16777603.

[20] Zheng GX, et al. Massively parallel digital transcriptional profiling of single cells. *Nat Commun* 8:14049, 2017. PMID 28091601.